

Go to Jenkins

<https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.118/bin/apache-tomcat-9.0.118.zip>

create a new job

give item name

select : pipeline

Fork : <https://github.com/akshu20791/addressbook-cicd-project>

```
pipeline{
  agent any
  stages{
    stage("clone from github"){
      steps{
        git 'https://github.com/akshu20791/addressbook-cicd-project/'
        echo 'clone done'
      }
    }
    stage("compile the project"){
      steps{
        sh 'mvn compile'
      }
    }
    stage("run the test cases"){
      steps{
        sh 'mvn test'
      }
    }
    stage("package the project"){
```

```
    steps{  
      sh 'mvn package'  
    }  
  }  
}
```

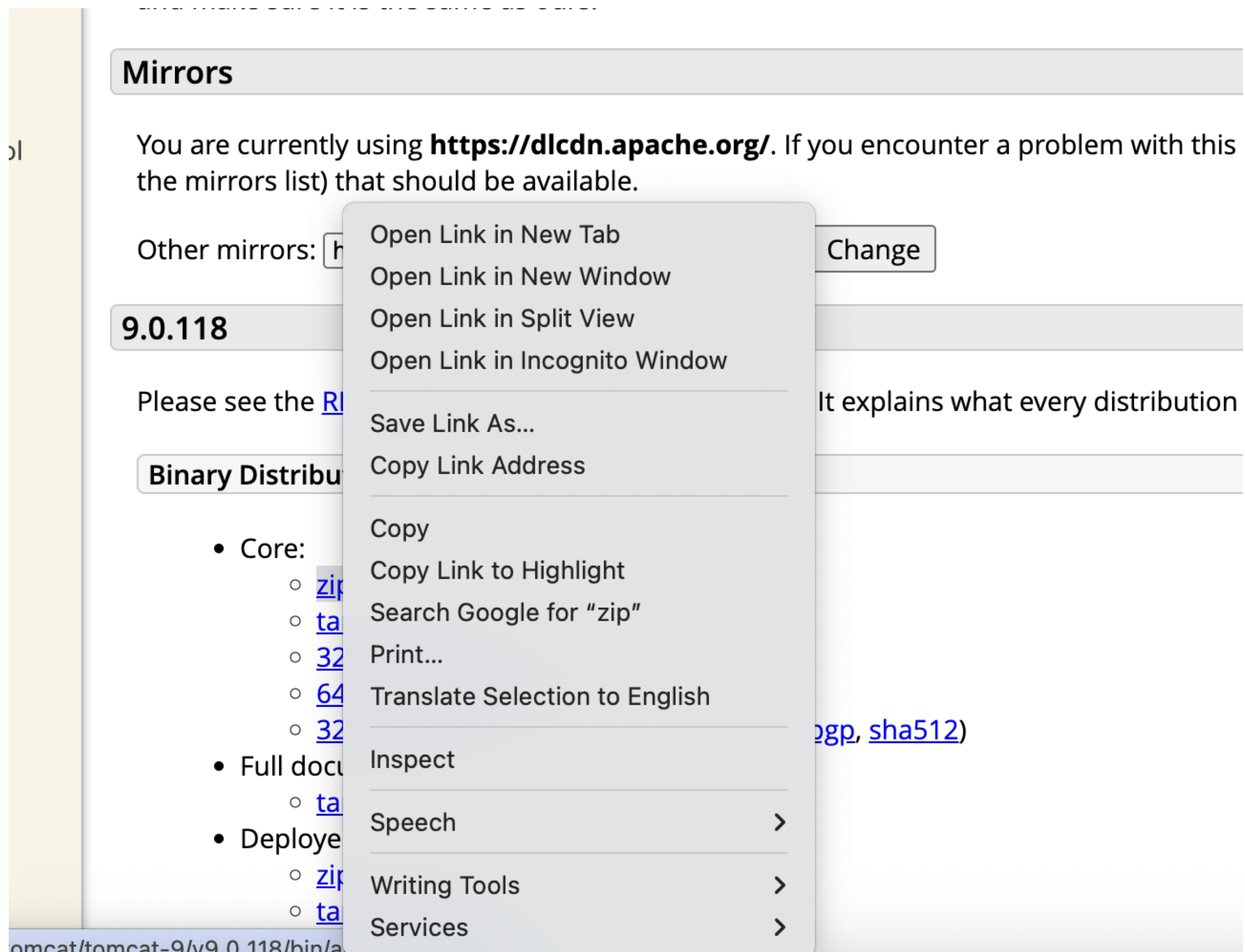
Now we need to deploy the app

for that we need to use TOMCAT

which is webserver required to deploy the java based app

<https://tomcat.apache.org/download-90.cgi>

Right click on zip and copy address



wget

<https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.118/bin/apache-tomcat-9.0.118.zip>

apt install unzip -y

unzip [apache-tomcat-9.0.118.zip](https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.118/bin/apache-tomcat-9.0.118.zip)

So jenkins and tomcat works on port 8080...we cannot run two different software/app in same port

cd apache-tomcat-9.0.118

ls

cd conf

ls

Here you will see server.xml file

vi server.xml

Replace 8080 with 9090

```
and responses are returned. Documentation at :  
Java HTTP Connector: /docs/config/http.html  
Java AJP Connector: /docs/config/ajp.html  
APR (HTTP/AJP) Connector: /docs/apr.html  
Define a non-SSL/TLS HTTP/1.1 Connector on port 8080  
-->  
<Connector port="8080" protocol="HTTP/1.1"  
    connectionTimeout="20000"  
    redirectPort="8443"  
    maxParameterCount="1000"  
/>  
  
<!-- A "Connector" using the shared thread pool-->  
<!--  
<Connector executor="tomcatThreadPool"  
    port="8080" protocol="HTTP/1.1"
```

i-060d962cf05613bba (jenkins-machine)

```

Java HTTP Connector: /docs/config/http.html
Java AJP Connector: /docs/config/ajp.html
APR (HTTP/AJP) Connector: /docs/apr.html
Define a non-SSL/TLS HTTP/1.1 Connector on port 8080
-->
<Connector port="9090" protocol="HTTP/1.1"
    connectionTimeout="20000"
    redirectPort="8443"
    maxParameterCount="1000"
/>
<!-- A "Connector" using the shared thread pool-->
<!--
<Connector executor="tomcatThreadPool"
    port="8080" protocol="HTTP/1.1"
-- INSERT --

```

cd ..

cd bin

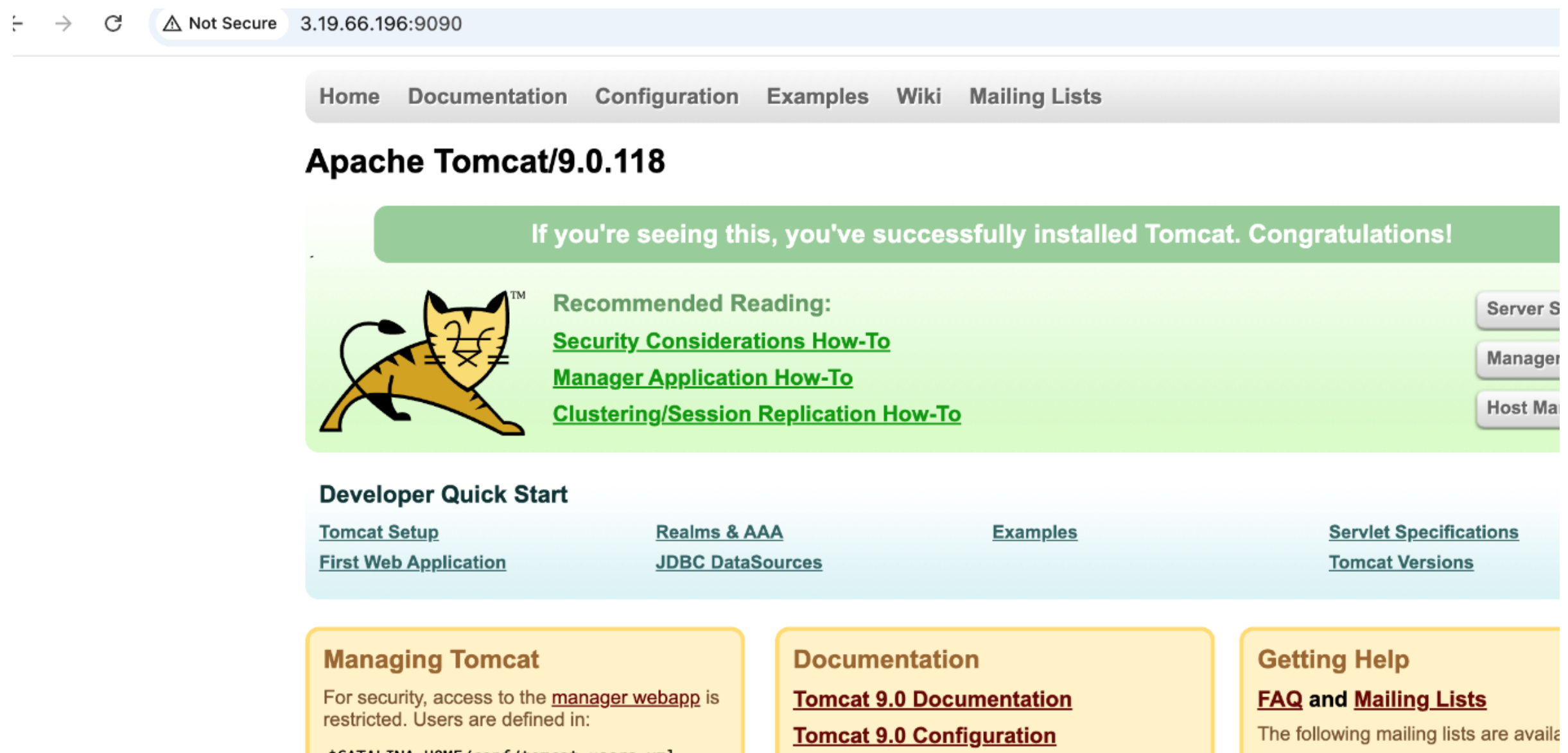
you need to give read write and execute permission to all the files ending with .sh

chmod +x *.sh

./startup.sh

your tomcat will not run on port 9090

publicip:9090



cd ..

To deploy java app ...what ever package (jar or war) is created we need to copy it to webapps directory inside the tomcat folder

```
root@ip-172-31-46-82:/home/ubuntu/apache-tomcat-9.0.118# ls
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
root@ip-172-31-46-82:/home/ubuntu/apache-tomcat-9.0.118#
```

Lets first check where the package is created

lets go to jenkins...lets build the job

 **Jenkins** / myfirstprojectdeployment

Status

</> Changes

▶ Build Now

⚙️ Configure

🗑️ Delete Pipeline

📁 Stages

✎ Rename

❓ Pipeline Syntax

✔️ myfirstprojectdeployment

Permalinks

- [Last build \(#3\), 21 min ago](#)
- [Last stable build \(#3\), 21 min ago](#)
- [Last successful build \(#3\), 21 min ago](#)
- [Last completed build \(#3\), 21 min ago](#)

Builds > ⋮

🔍 Filter /

Today

✔️ #4 04:46 ✖ ▼

✔️ #3 04:25 ▼

✔️ #2 04:15 ▼

✔️ #1 04:13 ▼

click on your latest build..if success

clcik on console output

Public

```
Jenkins / myfirstprojectdeployment / #4

Cause:at com.edurekademo.utilities.ExceptionThrower.doCheck(ExceptionThrower.java:43)
Cause:at com.edurekademo.utilities.TestLogger.testGetErrorMessage5(TestLogger.java:75)
[pool-1-thread-8] INFO com.edurekademo.utilities.TestLogger - java.io.IOException: TESTIOEXCEPTION
Cause:at com.edurekademo.utilities.ExceptionThrower.doNothing(ExceptionThrower.java:18)
Cause:at com.edurekademo.utilities.TestLogger.testGetErrorMessage4(TestLogger.java:60)
Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.065 sec - in com.edurekademo.utilities.TestLog
Running com.edurekademo.utilities.TestGenericComparator
Tests run: 14, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.053 sec - in
com.edurekademo.utilities.TestGenericComparator

Results :

Tests run: 23, Failures: 0, Errors: 0, Skipped: 0

[ [1;34mINFO [m]
[ [1;34mINFO [m] [1m--- [0;32mwar:3.3.2:war [m [1m(default-war) [m @ [36maddressbook [0;1m --- [m
[ [1;34mINFO [m] Packaging webapp
[ [1;34mINFO [m] Assembling webapp [addressbook] in [/var/lib/jenkins/workspace/myfirstprojectdeployment/target/a
[ [1;34mINFO [m] Processing war project
[ [1;34mINFO [m] Building war: /var/lib/jenkins/workspace/myfirstprojectdeployment/target/addressbook.war
[ [1;34mINFO [m] [1m----- [m
[ [1;34mINFO [m] [1;32mBUILD SUCCESS [m
[ [1;34mINFO [m] [1m----- [m
[ [1;34mINFO [m] Total time: 5.274 s
[ [1;34mINFO [m] Finished at: 2026-06-09T04:46:42Z
[ [1;34mINFO [m] [1m----- [m
[Pipeline] }
```

in logs you will find this..this is basically the war file created by jenkins

copy the location

put it in notepad

now go back to your machine and check the location of webapps directory

cd webapps

pwd

```
Basic login: Tue Jun 9 04:55:04 2026 from 3.16.146.5
ubuntu@ip-172-31-46-82:~$ sudo su
root@ip-172-31-46-82:/home/ubuntu# ls
apache-tomcat-9.0.118  apache-tomcat-9.0.118.zip
root@ip-172-31-46-82:/home/ubuntu# cd apache-tomcat-9.0.118
root@ip-172-31-46-82:/home/ubuntu/apache-tomcat-9.0.118# apache-tomcat-9.0.118
apache-tomcat-9.0.118: command not found
root@ip-172-31-46-82:/home/ubuntu/apache-tomcat-9.0.118# cd webapps
root@ip-172-31-46-82:/home/ubuntu/apache-tomcat-9.0.118/webapps# ls
ROOT  docs  examples  host-manager  manager
root@ip-172-31-46-82:/home/ubuntu/apache-tomcat-9.0.118/webapps# pwd
/home/ubuntu/apache-tomcat-9.0.118/webapps
root@ip-172-31-46-82:/home/ubuntu/apache-tomcat-9.0.118/webapps#
```

copy the location

/home/ubuntu/apache-tomcat-9.0.118/webapps

Go to jenkins --- go to your pipeline -- configure

The screenshot shows the Jenkins web interface in a browser. The address bar indicates the URL is `3.19.66.196:8080/job/myfirstprojectdeployment/`. The page title is "Jenkins / myfirstprojectdeployment". On the left sidebar, the "Configure" option is highlighted with a yellow background. The main content area shows a green checkmark icon and the pipeline name "myfirstprojectdeployment". Below this, there is a section titled "Permalinks" with a list of links for the last build, last stable build, last successful build, and last completed build, all of which are build #4 and completed 5 minutes and 11 seconds ago. At the bottom, there is a "Builds" section with a search filter and a list of builds. The builds are listed as follows:

Build Number	Duration
#4	04:46
#3	04:25
#2	04:15
#1	04:13

lets now complete the pipeline

We will add up the deploying stage where in the steps we will mention:

sudo cp source location of war file destination

```
pipeline{
  agent any
  stages{
    stage("clone from github"){
      steps{
        git 'https://github.com/akshu20791/addressbook-cicd-project/'
        echo 'clone done'
      }
    }
    stage("compile the project"){
      steps{
        sh 'mvn compile'
      }
    }
    stage("run the test cases"){
      steps{
        sh 'mvn test'
      }
    }
    stage("package the project"){
      steps{
        sh 'mvn package'
      }
    }
    stage("deploying the project to tomcat"){
      steps{
        sh "sudo cp
/var/lib/jenkins/workspace/myfirstprojectdeployment/target/addressbook.war
/home/ubuntu/apache-tomcat-9.0.118/webapps"
```

```
    }  
  }  
}  
  
}
```

You have to change the location of your source and destination

Build now

you will get the error !!!

This error says that your jenkins user does not have sudo permission so copy will not work...we need to give admin permission to jenkins

 **Jenkins** / myfirstprojectdeployment

Status

Changes

Build Now

Configure

Delete Pipeline

Stages

Rename

Pipeline Syntax

Builds >

Filter

Today

✖

#5

04:54

▼

✓

#4

04:46

▼

✓

#3

04:25

▼

✓

#2

04:15

▼

✓

#1

04:13

▼

✓ myfirstprojectdeployment

Permalinks

Last build (#4), 8 min 6 sec ago

Last stable build (#4), 8 min 6 sec ago

Last successful build (#4), 8 min 6 sec ago

Last completed build (#4), 8 min 6 sec ago

Prev

Click on error build

go to console output

```
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (deploying the project to tomcat)
[Pipeline] sh
+ sudo cp /var/lib/jenkins/workspace/myfirstprojectdeployment/target/addressbook.war /home/ubuntu/apache-tomcat-
9.0.118/webapps
sudo: I'm sorry jenkins. I'm afraid I can't do that
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
ERROR: script returned exit code 1
Finished: FAILURE
```

because jenkins does not have root permission

We will now give the root permission to jenkins ...

Go to machine ...there is a sudoers file...

visudo

jenkins ALL=(ALL:ALL) NOPASSWD: ALL

(put this below root)

```
# Ditto for GPG agent
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL
jenkins ALL=(ALL:ALL) NOPASSWD: ALL
# Members of the admin group may gain root privileges
%admin  ALL=(ALL) ALL

# Allow members of group sudo to execute any command

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify

i-060d962cf05613bba (jenkins-machine)
Public IP: 3.10.66.100   Private IP: 172.31.46.22
```

ctrl x -> y -> enter to come out of the file

For this permission to take place we need to restart jenkins

service jenkins restart

```
root@ip-172-31-46-82:/home/ubuntu/apache-tomcat-9.0.118/webapps# visudo
root@ip-172-31-46-82:/home/ubuntu/apache-tomcat-9.0.118/webapps# service jenkins restart
```

you need to relogin the jenkins from browser

build the job again

Go to your job -> build now

Status

</> Changes

Build Now

Configure

Delete Pipeline

Stages

Rename

Pipeline Syntax

Builds >

Filter

Today

#6 04:59

#5 04:54

#4 04:46

#3 04:25

#2 04:15

#1 04:13

myfirstprojectdeployr

Permalinks

Last build (#5), 4 min 28 sec ago

Last stable build (#4), 12 min ago

Last successful build (#4), 12 min ago

Last failed build (#5), 4 min 28 sec ago

Last unsuccessful build (#5), 4 min 28 sec ago

Last completed build (#5), 4 min 28 sec ago

Wait for it to be success

if you get error you can put the logs in chatgpt and try to identify issuer

Public

build is success

 **Jenkins** / myfirstprojectdeployment

Status

Changes

Build Now

Configure

Delete Pipeline

Stages

Rename

Pipeline Syntax

Builds >

Filter

Today

✓

#6

04:59

▼

✗

#5

04:54

▼

✓

#4

04:46

▼

✓

#3

04:25

▼

✗

 myfirstprojectdeploymer

Permalinks

Last build (#5), 4 min 28 sec ago

Last stable build (#4), 12 min ago

Last successful build (#4), 12 min ago

Last failed build (#5), 4 min 28 sec ago

Last unsuccessful build (#5), 4 min 28 sec ago

Last completed build (#5), 4 min 28 sec ago

let see if app is running:

Public

Not Secure 3.19.66.196:9090/addressbook/

Iter contacts...		
First Name	Last Name	Email
George	White	george@white.com
Daniel	Thompson	daniel@thompson.com
Timothy	Jones	timothy@jones.com
Peter	Wilson	peter@wilson.com
Dan	Robinson	dan@robinson.com
Dan	Davis	dan@davis.com
Olivia	Davis	olivia@davis.com
Dan	Smith	dan@smith.com
Daniel	Anderson	daniel@anderson.com
Alice	Thomas	alice@thomas.com
Linda	Harris	linda@harris.com
Daniel	Robinson	daniel@robinson.com
Mike	Young	mike@young.com

<http://publicipofmachine:9090/addressbook>

in my case: <http://3.19.66.196:9090/addressbook/>